

Banking management system

DBMS Assignment



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BSCS-4C

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Database Design & Implementation

1. Logical Schema Mapping

- Convert your EER diagram into a relational schema.
- Clearly identify **Primary Keys (PK)** and **Foreign Keys (FK)**.
- Clearly show mapping of **Supertype/Subtype relationships**.

Table	Primary Key	Non-Key Attributes	Foreign Keys
Customer	CustomerID (PK)	FullName, CNIC, ContactNo, HouseNo, Street, City, PostalCode, DateOfBirth, RegistrationDate	—
Branch	BranchID (PK)	BranchName, City, PostalCode, ContactNo	ManagerID → Employee(EmployeeID)
Employee	EmployeeID (PK)	FirstName, LastName, Designation, Salary, Phone, Email, JoiningDate	BranchID → Branch(BranchID)
Account	AccountID (PK)	AccountType, Balance, Status, OpeningDate	CustomerID → Customer(CustomerID)
SavingsAccount	AccountID (PK, FK)	InterestRate	AccountID → Account(AccountID)
CurrentAccount	AccountID (PK, FK)	OverdraftLimit	AccountID → Account(AccountID)
Loan	LoanID (PK)	LoanType, ApprovedAmount, InterestRate, RepaymentSchedule, OutstandingBalance, LoanStatus	CustomerID → Customer(CustomerID), ApprovedByEmpID → Employee(EmployeeID)
Transaction	TransactionID (PK)	Type, Amount, Timestamp, Status	AccountID → Account(AccountID), ProcessedByEmpID → Employee(EmployeeID)
ATM	ATMID (PK)	Location, CashAvailability, OperationalStatus	BranchID → Branch(BranchID)

Supertype / Subtype Mapping

- Account stores all common attributes with AccountType.
- SavingsAccount stores InterestRate, with AccountID as both PK and FK referencing Account.
- CurrentAccount stores OverdraftLimit, with AccountID as both PK and FK referencing Account.
- Every Savings account row must have a matching SavingsAccount row, and every Current account row must have a matching CurrentAccount row.

Relational Schema:

Customer (CustomerID,
FullName,
CNIC,
ContactNo,
HouseNo,
Street,
City,
PostalCode,
DateOfBirth,
RegistrationDate
)

Branch (
BranchID,
BranchName,
City,
PostalCode,
ContactNo,
ManagerID
)

Employee (EmployeeID,
FirstName,
LastName,
Designation,
Salary,
Phone,
Email,
JoiningDate,
BranchID
)

Account (AccountID,
AccountType,
Balance,
Status,
OpeningDate,
CustomerID)

)

SavingsAccount (AccountID,
InterestRate
)

CurrentAccount (AccountID,
OverdraftLimit
)

Loan (LoanID,
LoanType,
ApprovedAmount,
InterestRate,
RepaymentSchedule,
OutstandingBalance,
LoanStatus,
CustomerID,
ApprovedByEmpID
)

Transaction (TransactionID,
Type,
Amount,
Timestamp,
Status,
AccountID,
ProcessedByEmpID
)

ATM (ATMID,
Location,
CashAvailability,
OperationalStatus,
BranchID
)

2. Database Physical Construction (DDL)

- Write and execute SQL DDL statements to create tables
- Apply constraints: **NOT NULL**, **UNIQUE**, **CHECK** etc.
- Ensure proper **referential integrity** using foreign keys.

```
-- Customer
CREATE TABLE Customer(
    CustomerID INT PRIMARY KEY IDENTITY(1,1),
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50) NOT NULL,
    CNIC CHAR(15) NOT NULL UNIQUE
    CHECK (CNIC LIKE '[0-9][0-9][0-9][0-9][0-9]-[0-9][0-9][0-9][0-9][0-9][0-9][0-9]-[0-9]'),
    ContactNo VARCHAR(20) NOT NULL,
    HouseNo VARCHAR(20) NULL,
    Street VARCHAR(80) NULL,
    City VARCHAR(80) NOT NULL,
    PostalCode VARCHAR(10) NULL,
    DateOfBirth DATE NOT NULL,
    RegistrationDate DATE NOT NULL DEFAULT GETDATE()
);

--Branch
CREATE TABLE Branch(
    BranchID INT PRIMARY KEY IDENTITY(1,1),
    BranchName VARCHAR(100) NOT NULL,
    City VARCHAR(80) NOT NULL,
    PostalCode VARCHAR(10) NOT NULL,
    ContactNo VARCHAR(20) NOT NULL UNIQUE,
    ManagerID INT NULL
);

--Employee
CREATE TABLE Employee(
    EmployeeID INT PRIMARY KEY IDENTITY(1,1),
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50) NOT NULL,
    Designation VARCHAR(60) NOT NULL,
    Salary DECIMAL(12,2) NOT NULL CHECK (Salary > 0),
    Phone VARCHAR(20) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    JoiningDate DATE NOT NULL,
    BranchID INT NOT NULL,
    FOREIGN KEY (BranchID) REFERENCES Branch(BranchID)
);

-- Adding Constraint
ALTER TABLE Branch
    ADD CONSTRAINT FK_Branch_Manager
    FOREIGN KEY (ManagerID) REFERENCES Employee(EmployeeID);
```

```

-- Account (supertype)
CREATE TABLE Account(
    AccountID INT PRIMARY KEY IDENTITY(1,1),
    AccountType VARCHAR(10) NOT NULL CHECK (AccountType IN ('Savings', 'Current')),
    Balance DECIMAL(15,2) NOT NULL DEFAULT 0.00
    CHECK (Balance >= 0),
    Status VARCHAR(10) NOT NULL DEFAULT 'Active'
    CHECK (Status IN ('Active', 'Frozen', 'Closed')),
    OpeningDate DATE NOT NULL DEFAULT GETDATE(),
    CustomerID INT NOT NULL,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID)
);

-- Savings Account (subtype)
CREATE TABLE SavingsAccount(
    AccountID INT PRIMARY KEY,
    InterestRate DECIMAL(5,2) NOT NULL CHECK (InterestRate >= 0),
    FOREIGN KEY (AccountID) REFERENCES Account(AccountID)
);

-- Current Account (subtype)
CREATE TABLE CurrentAccount(
    AccountID INT PRIMARY KEY,
    OverdraftLimit DECIMAL(12,2) NOT NULL DEFAULT 0.00 CHECK (OverdraftLimit >= 0),
    FOREIGN KEY (AccountID) REFERENCES Account(AccountID)
);

-- Loan
CREATE TABLE Loan(
    LoanID INT PRIMARY KEY IDENTITY(1,1),
    LoanType VARCHAR(20) NOT NULL
    CHECK (LoanType IN ('Personal', 'Home', 'Car', 'Business')),
    ApprovedAmount DECIMAL(15,2) NOT NULL CHECK (ApprovedAmount > 0),
    InterestRate DECIMAL(5,2) NOT NULL CHECK (InterestRate >= 0),
    RepaymentSchedule VARCHAR(20) NOT NULL DEFAULT 'Monthly',
    OutstandingBalance AS (ApprovedAmount),
    LoanStatus VARCHAR(10) NOT NULL DEFAULT 'Pending'
    CHECK (LoanStatus IN ('Pending', 'Approved', 'Closed')),
    CustomerID INT NOT NULL,
    ApprovedByEmpID INT NULL,
    FOREIGN KEY (CustomerID) REFERENCES Customer(CustomerID),
    FOREIGN KEY (ApprovedByEmpID) REFERENCES Employee(EmployeeID)
);

--Transaction
CREATE TABLE [Transaction](
    TransactionID INT PRIMARY KEY IDENTITY(1,1),
    Type VARCHAR(15) NOT NULL
    CHECK (Type IN ('Deposit', 'Withdrawal', 'Transfer')),
    Amount DECIMAL(15,2) NOT NULL CHECK (Amount > 0),
    Timestamp DATETIME NOT NULL DEFAULT GETDATE(),
    Status VARCHAR(10) NOT NULL DEFAULT 'Pending'
    CHECK (Status IN ('Completed', 'Failed', 'Pending')),
    AccountID INT NOT NULL,
    ProcessedByEmpID INT NOT NULL,
    FOREIGN KEY (AccountID) REFERENCES Account(AccountID),
    FOREIGN KEY (ProcessedByEmpID) REFERENCES Employee(EmployeeID)
);

```

```
-- ATM
CREATE TABLE ATM(
  ATMID INT PRIMARY KEY IDENTITY(1,1),
  Location VARCHAR(150) NOT NULL,
  CashAvailability DECIMAL(15,2) NOT NULL CHECK (CashAvailability >= 0),
  OperationalStatus VARCHAR(20) NOT NULL DEFAULT 'Active'
  CHECK (OperationalStatus IN ('Active', 'Under Maintenance', 'Offline')),
  BranchID INT NOT NULL,
  FOREIGN KEY (BranchID) REFERENCES Branch(BranchID)
);
```

The following screenshots show all the tables have successfully been created.

Results Messages											
CustomerID	FirstName	LastName	CNIC	ContactNo	HouseNo	Street	City	PostalCode	DateOfBirth	RegistrationDate	
BranchID	BranchName	City	PostalCode	ContactNo	ManagerID						
EmployeeID	FirstName	LastName	Designation	Salary	Phone	Email	JoiningDate	BranchID			
AccountID	AccountType	Balance	Status	OpeningDate	CustomerID						
AccountID	InterestRate										
AccountID	OverdraftLimit										
LoanID	LoanType	ApprovedAmount	InterestRate	RepaymentSchedule	OutstandingBalance	LoanStatus	CustomerID	ApprovedByEmpID			
TransactionID	Type	Amount	Timestamp	Status	AccountID	ProcessedByEmpID					
ATMID	Location	CashAvailability	OperationalStatus	BranchID							

3. Data Population (DML)

- Insert sample data into all tables.
- Each table must contain **10–15 meaningful and unique records**.

Inserted Values:

```
-- Branch(without manager)
INSERT INTO Branch (BranchName, City, PostalCode, ContactNo)
VALUES('Blue Area Branch','Islamabad','44000','051-1112222'),
('Saddar Branch','Rawalpindi','46000','051-3334444'),
('Gulberg Branch','Lahore','54000','042-5556666'),
('Clifton Branch','Karachi','75600','021-7778888'),
('Cantt Branch','Peshawar','25000','091-9990000'),
('DHA Phase 1 Branch','Lahore','54792','042-1112233'),
('Johar Town Branch','Lahore','54782','042-4445566'),
('Gulshan Branch','Karachi','75300','021-2223344'),
('Hayatabad Branch','Peshawar','25100','091-5556677'),
('F-7 Markaz Branch','Islamabad','44220','051-7778899');

-- Employee
INSERT INTO Employee (FirstName, LastName, Designation, Salary, Phone, Email, JoiningDate, BranchID)
VALUES
('Amir','Khan','Branch Manager',120000,'0300-1112222','amir.khan@bank.pk','2018-03-15',1),
('Sara','Ahmed','Loan Officer',85000,'0301-2223333','sara.ahmed@bank.pk','2019-07-01',1),
('Bilal','Raza','Branch Manager',115000,'0302-3334444','bilal.raza@bank.pk','2017-11-20',2),
('Nadia','Malik','Teller',55000,'0303-4445555','nadia.malik@bank.pk','2021-01-10',2),
('Usman','Ali','Branch Manager',125000,'0304-5556666','usman.ali@bank.pk','2016-06-05',3),
('Hira','Shah','Loan Officer',88000,'0305-6667777','hira.shah@bank.pk','2020-09-15',3),
('Kamran','Siddiqui','Branch Manager',110000,'0306-7778888','kamran.s@bank.pk','2015-04-22',4),
('Zainab','Hassan','Teller',52000,'0307-8889999','zainab.h@bank.pk','2022-03-01',4),
('Tariq','Mahmood','Branch Manager',118000,'0308-9990000','tariq.m@bank.pk','2017-08-18',5),
('Ayesha','Farooq','Loan Officer',82000,'0309-0001111','ayesha.f@bank.pk','2021-11-05',5),
('Imran','Yusuf','Teller',54000,'0310-1112222','imran.y@bank.pk','2020-05-20',6),
('Fatima','Qureshi','Teller',53000,'0311-2223333','fatima.q@bank.pk','2019-12-10',7);

-- Update ManagerID in branch
UPDATE Branch SET ManagerID=1 WHERE BranchID=1;
UPDATE Branch SET ManagerID=3 WHERE BranchID=2;
UPDATE Branch SET ManagerID=5 WHERE BranchID=3;
UPDATE Branch SET ManagerID=7 WHERE BranchID=4;
UPDATE Branch SET ManagerID=9 WHERE BranchID=5;
UPDATE Branch SET ManagerID=1 WHERE BranchID=6;
UPDATE Branch SET ManagerID=3 WHERE BranchID=7;
UPDATE Branch SET ManagerID=5 WHERE BranchID=8;
UPDATE Branch SET ManagerID=7 WHERE BranchID=9;
UPDATE Branch SET ManagerID=9 WHERE BranchID=10;

-- Customer
INSERT INTO Customer (FirstName,LastName,CNIC,ContactNo,HouseNo,Street,City,PostalCode,DateOfBirth)
VALUES
('Ali','Raza','35202-1234567-1','0300-1111111','12','Khayaban-e-Iqbal','Islamabad','44000','1990-05-10'),
('Sana','Malik','35202-2345678-2','0301-2222222','45','Jinnah Avenue','Islamabad','44000','1985-08-22'),
('Hassan','Ahmed','37405-3456789-3','0302-3333333','7','Mall Road','Rawalpindi','46000','1992-03-15'),
('Maryam','Khan','35201-4567890-4','0303-4444444','23','Allama Iqbal Rd','Lahore','54000','1988-11-30'),
('Fahad','Siddiqui','21303-5678901-5','0304-5555555','89','Clifton Block 5','Karachi','75600','1995-07-07'),
('Amna','Hussain','17301-6789012-6','0305-6666666','3','University Rd','Peshawar','25000','1993-01-25'),
('Umar','Farooq','35202-7890123-7','0306-7777777','56','F-8 Markaz','Islamabad','44220','1987-09-14'),
('Saira','Iqbal','35401-8901234-8','0307-8888888','14','DHA Phase 2','Lahore','54792','1991-04-03'),
('Junaid','Baig','21301-9012345-9','0308-9999999','67','Gulshan Block 9','Karachi','75300','1986-12-20'),
('Rabia','Shah','17302-0123456-0','0309-0000000','9','Hayatabad Ph 3','Peshawar','25100','1994-06-18'),
('Zubair','Mirza','35202-1357924-1','0310-1010101','31','G-10/2','Islamabad','44000','1989-02-28'),
('Hina','Javed','35201-2468013-2','0311-2020202','18','Johar Town C-Blk','Lahore','54782','1996-10-05');
```

```

-- Account (supertype)
INSERT INTO Account (AccountType, Balance, Status, OpeningDate, CustomerID)
VALUES
('Savings', 150000.00, 'Active', '2020-01-15', 1),
('Current', 500000.00, 'Active', '2019-06-01', 2),
('Savings', 75000.00, 'Active', '2021-03-20', 3),
('Current', 250000.00, 'Frozen', '2018-11-10', 4),
('Savings', 320000.00, 'Active', '2022-07-05', 5),
('Savings', 45000.00, 'Active', '2020-09-12', 6),
('Current', 180000.00, 'Active', '2017-04-28', 7),
('Savings', 620000.00, 'Active', '2019-01-30', 8),
('Current', 95000.00, 'Closed', '2016-08-15', 9),
('Savings', 410000.00, 'Active', '2023-02-14', 10),
('Savings', 30000.00, 'Active', '2021-11-07', 11),
('Current', 780000.00, 'Active', '2020-05-23', 12);

-- SavingsAccount (subtype)
INSERT INTO SavingsAccount (AccountID, InterestRate)
VALUES
(1, 7.50), (3, 6.00), (5, 7.00), (6, 6.50), (8, 7.25), (10, 6.75), (11, 6.00);

-- CurrentAccount (subtype)
INSERT INTO CurrentAccount (AccountID, OverdraftLimit)
VALUES
(2, 100000.00), (4, 50000.00), (7, 75000.00), (9, 30000.00), (12, 200000.00);

-- Loan
INSERT INTO Loan (LoanType, ApprovedAmount, InterestRate, RepaymentSchedule,
LoanStatus, CustomerID, ApprovedByEmpID)
VALUES
('Personal', 500000, 12.50, 'Monthly', 'Approved', 1, 2),
('Home', 5000000, 10.00, 'Monthly', 'Approved', 2, 6),
('Car', 800000, 11.00, 'Monthly', 'Pending', 3, NULL),
('Business', 2000000, 9.50, 'Monthly', 'Approved', 4, 2),
('Personal', 300000, 13.00, 'Monthly', 'Approved', 5, 10),
('Home', 3500000, 10.50, 'Monthly', 'Pending', 6, NULL),
('Car', 600000, 11.50, 'Monthly', 'Approved', 7, 6),
('Business', 1500000, 9.00, 'Monthly', 'Closed', 8, 10),
('Personal', 200000, 14.00, 'Monthly', 'Approved', 9, 2),
('Home', 4000000, 10.25, 'Monthly', 'Approved', 10, 6),
('Car', 750000, 11.75, 'Monthly', 'Pending', 11, NULL),
('Business', 2500000, 8.75, 'Monthly', 'Approved', 12, 10);

```

```

-- Transaction
INSERT INTO [Transaction] (Type,Amount,Timestamp,Status,AccountID,ProcessedByEmpID)
VALUES
('Deposit',50000,'2024-01-05 09:15:00','Completed',1,4),
('Withdrawal',20000,'2024-01-06 10:30:00','Completed',1,4),
('Transfer',30000,'2024-01-07 11:00:00','Completed',2,8),
('Deposit',100000,'2024-01-08 09:00:00','Completed',2,8),
('Withdrawal',15000,'2024-01-09 14:20:00','Failed',3,4),
('Deposit',75000,'2024-01-10 10:45:00','Completed',5,8),
('Transfer',25000,'2024-01-11 13:00:00','Completed',6,11),
('Deposit',200000,'2024-01-12 09:30:00','Completed',7,11),
('Withdrawal',50000,'2024-01-13 15:10:00','Completed',8,12),
('Deposit',80000,'2024-01-14 11:20:00','Completed',10,12),
('Transfer',40000,'2024-01-15 08:55:00','Pending',11,4),
('Deposit',300000,'2024-01-16 12:00:00','Completed',12,8),
('Withdrawal',10000,'2024-01-17 16:30:00','Completed',1,4),
('Deposit',60000,'2024-01-18 09:45:00','Completed',3,11),
('Transfer',90000,'2024-01-19 14:00:00','Completed',5,12);

-- ATM
INSERT INTO ATM (Location, CashAvailability, OperationalStatus, BranchID)
VALUES
('Blue Area, Jinnah Ave, Islamabad',2000000,'Active',1),
('Centaurus Mall, Islamabad',1500000,'Active',1),
('Saddar Bazaar, Rawalpindi',1800000,'Active',2),
('Gulberg Main Market, Lahore',2500000,'Active',3),
('Clifton Bridge, Karachi',500000,'Under Maintenance',4),
('Peshawar Cantt Main Gate',1200000,'Active',5),
('DHA Phase 1 Commercial, Lahore',1700000,'Active',6),
('Johar Town Al-Fatah, Lahore',300000,'Active',7),
('Gulshan-e-Iqbal, Karachi',2200000,'Active',8),
('Hayatabad Phase 3, Peshawar',900000,'Offline',9),
('F-7 Markaz, Islamabad',1600000,'Active',10),
('University Road, Peshawar',1100000,'Active',5);

```

Tables:

- Customer

	CustomerID	FirstName	LastName	CNIC	ContactNo	HouseNo	Street	City	PostalCode	DateOfBirth	RegistrationDate
1	1	Ali	Raza	35202-1234567-1	0300-1111111	12	Khayaban-e-Iqbal	Islamabad	44000	1990-05-10	2026-05-10
2	2	Sana	Malik	35202-2345678-2	0301-2222222	45	Jinnah Avenue	Islamabad	44000	1985-08-22	2026-05-10
3	3	Hassan	Ahmed	37405-3456789-3	0302-3333333	7	Mall Road	Rawalpindi	46000	1992-03-15	2026-05-10
4	4	Maryam	Khan	35201-4567890-4	0303-4444444	23	Allama Iqbal Rd	Lahore	54000	1988-11-30	2026-05-10
5	5	Fahad	Siddiqui	21303-5678901-5	0304-5555555	89	Clifton Block 5	Karachi	75600	1995-07-07	2026-05-10
6	6	Amna	Hussain	17301-6789012-6	0305-6666666	3	University Rd	Peshawar	25000	1993-01-25	2026-05-10
7	7	Umar	Farooq	35202-7890123-7	0306-7777777	56	F-8 Markaz	Islamabad	44220	1987-09-14	2026-05-10
8	8	Saira	Iqbal	35401-8901234-8	0307-8888888	14	DHA Phase 2	Lahore	54792	1991-04-03	2026-05-10
9	9	Junaid	Baig	21301-9012345-9	0308-9999999	67	Gulshan Block 9	Karachi	75300	1986-12-20	2026-05-10
10	10	Rabia	Shah	17302-0123456-0	0309-0000000	9	Hayatabad Ph 3	Peshawar	25100	1994-06-18	2026-05-10
11	11	Zubair	Mirza	35202-1357924-1	0310-1010101	31	G-10/2	Islamabad	44000	1989-02-28	2026-05-10
12	12	Hina	Javed	35201-2468013-2	0311-2020202	18	Johar Town C-Blk	Lahore	54782	1996-10-05	2026-05-10

- Branch

	BranchID	BranchName	City	PostalCode	ContactNo	ManagerID
1	1	Blue Area Branch	Islamabad	44000	051-1112222	1
2	2	Saddar Branch	Rawalpindi	46000	051-3334444	3
3	3	Gulberg Branch	Lahore	54000	042-5556666	5
4	4	Clifton Branch	Karachi	75600	021-7778888	7
5	5	Cantt Branch	Peshawar	25000	091-9990000	9
6	6	DHA Phase 1 Branch	Lahore	54792	042-1112233	1
7	7	Johar Town Branch	Lahore	54782	042-4445566	3
8	8	Gulshan Branch	Karachi	75300	021-2223344	5
9	9	Hayatabad Branch	Peshawar	25100	091-5556677	7
10	10	F-7 Markaz Branch	Islamabad	44220	051-7778899	9

- Employee

	EmployeeID	FirstName	LastName	Designation	Salary	Phone	Email	JoiningDate	BranchID
1	1	Amir	Khan	Branch Manager	120000.00	0300-1112222	amir.khan@bank.pk	2018-03-15	1
2	2	Sara	Ahmed	Loan Officer	85000.00	0301-2223333	sara.ahmed@bank.pk	2019-07-01	1
3	3	Bilal	Raza	Branch Manager	115000.00	0302-3334444	bilal.raza@bank.pk	2017-11-20	2
4	4	Nadia	Malik	Teller	55000.00	0303-4445555	nadia.malik@bank.pk	2021-01-10	2
5	5	Usman	Ali	Branch Manager	125000.00	0304-5556666	usman.ali@bank.pk	2016-06-05	3
6	6	Hira	Shah	Loan Officer	88000.00	0305-6667777	hira.shah@bank.pk	2020-09-15	3
7	7	Kamran	Siddiqui	Branch Manager	110000.00	0306-7778888	kamran.s@bank.pk	2015-04-22	4
8	8	Zainab	Hassan	Teller	52000.00	0307-8889999	zainab.h@bank.pk	2022-03-01	4
9	9	Tariq	Mahmood	Branch Manager	118000.00	0308-9990000	tariq.m@bank.pk	2017-08-18	5
10	10	Ayesha	Farooq	Loan Officer	82000.00	0309-0001111	ayesha.f@bank.pk	2021-11-05	5
11	11	Imran	Yusuf	Teller	54000.00	0310-1112222	imran.y@bank.pk	2020-05-20	6
12	12	Fatima	Qureshi	Teller	53000.00	0311-2223333	fatima.q@bank.pk	2019-12-10	7

- Account

	AccountID	Account Type	Balance	Status	OpeningDate	CustomerID
1	1	Savings	150000.00	Active	2020-01-15	1
2	2	Current	500000.00	Active	2019-06-01	2
3	3	Savings	75000.00	Active	2021-03-20	3
4	4	Current	250000.00	Frozen	2018-11-10	4
5	5	Savings	320000.00	Active	2022-07-05	5
6	6	Savings	45000.00	Active	2020-09-12	6
7	7	Current	180000.00	Active	2017-04-28	7
8	8	Savings	620000.00	Active	2019-01-30	8
9	9	Current	95000.00	Closed	2016-08-15	9
10	10	Savings	410000.00	Active	2023-02-14	10
11	11	Savings	30000.00	Active	2021-11-07	11
12	12	Current	780000.00	Active	2020-05-23	12

- Savings Account

	AccountID	InterestRate
1	1	7.50
2	3	6.00
3	5	7.00
4	6	6.50
5	8	7.25
6	10	6.75
7	11	6.00

- Current Account

	AccountID	OverdraftLimit
1	2	100000.00
2	4	50000.00
3	7	75000.00
4	9	30000.00
5	12	200000.00

- Loan

	LoanID	Loan Type	ApprovedAmount	InterestRate	RepaymentSchedule	OutstandingBalance	LoanStatus	CustomerID	ApprovedByEmpID
1	1	Personal	500000.00	12.50	Monthly	500000.00	Approved	1	2
2	2	Home	5000000.00	10.00	Monthly	5000000.00	Approved	2	6
3	3	Car	800000.00	11.00	Monthly	800000.00	Pending	3	NULL
4	4	Business	2000000.00	9.50	Monthly	2000000.00	Approved	4	2
5	5	Personal	300000.00	13.00	Monthly	300000.00	Approved	5	10
6	6	Home	3500000.00	10.50	Monthly	3500000.00	Pending	6	NULL
7	7	Car	600000.00	11.50	Monthly	600000.00	Approved	7	6
8	8	Business	1500000.00	9.00	Monthly	1500000.00	Closed	8	10
9	9	Personal	200000.00	14.00	Monthly	200000.00	Approved	9	2
10	10	Home	4000000.00	10.25	Monthly	4000000.00	Approved	10	6
11	11	Car	750000.00	11.75	Monthly	750000.00	Pending	11	NULL
12	12	Business	2500000.00	8.75	Monthly	2500000.00	Approved	12	10

- Transaction

100 %

Results

Messages

	TransactionID	Type	Amount	Timestamp	Status	AccountID	ProcessedByEmpID
1	1	Deposit	50000.00	2024-01-05 09:15:00.000	Completed	1	4
2	2	Withdrawal	20000.00	2024-01-06 10:30:00.000	Completed	1	4
3	3	Transfer	30000.00	2024-01-07 11:00:00.000	Completed	2	8
4	4	Deposit	100000.00	2024-01-08 09:00:00.000	Completed	2	8
5	5	Withdrawal	15000.00	2024-01-09 14:20:00.000	Failed	3	4
6	6	Deposit	75000.00	2024-01-10 10:45:00.000	Completed	5	8
7	7	Transfer	25000.00	2024-01-11 13:00:00.000	Completed	6	11
8	8	Deposit	200000.00	2024-01-12 09:30:00.000	Completed	7	11
9	9	Withdrawal	50000.00	2024-01-13 15:10:00.000	Completed	8	12
10	10	Deposit	80000.00	2024-01-14 11:20:00.000	Completed	10	12
11	11	Transfer	40000.00	2024-01-15 08:55:00.000	Pending	11	4
12	12	Deposit	300000.00	2024-01-16 12:00:00.000	Completed	12	8
13	13	Withdrawal	10000.00	2024-01-17 16:30:00.000	Completed	1	4
14	14	Deposit	60000.00	2024-01-18 09:45:00.000	Completed	3	11
15	15	Transfer	90000.00	2024-01-19 14:00:00.000	Completed	5	12

- ATM

100 %

Results

Messages

	ATMID	Location	CashAvailability	OperationalStatus	BranchID
1	1	Blue Area, Jinnah Ave, Islamabad	2000000.00	Active	1
2	2	Centaurus Mall, Islamabad	1500000.00	Active	1
3	3	Saddar Bazaar, Rawalpindi	1800000.00	Active	2
4	4	Gulberg Main Market, Lahore	2500000.00	Active	3
5	5	Clifton Bridge, Karachi	500000.00	Under Maintenance	4
6	6	Peshawar Cantt Main Gate	1200000.00	Active	5
7	7	DHA Phase 1 Commercial, Lahore	1700000.00	Active	6
8	8	Johar Town Al-Fatah, Lahore	300000.00	Active	7
9	9	Gulshan-e-Iqbal, Karachi	2200000.00	Active	8
10	10	Hayatabad Phase 3, Peshawar	900000.00	Offline	9
11	11	F-7 Markaz, Islamabad	1600000.00	Active	10
12	12	University Road, Peshawar	1100000.00	Active	5

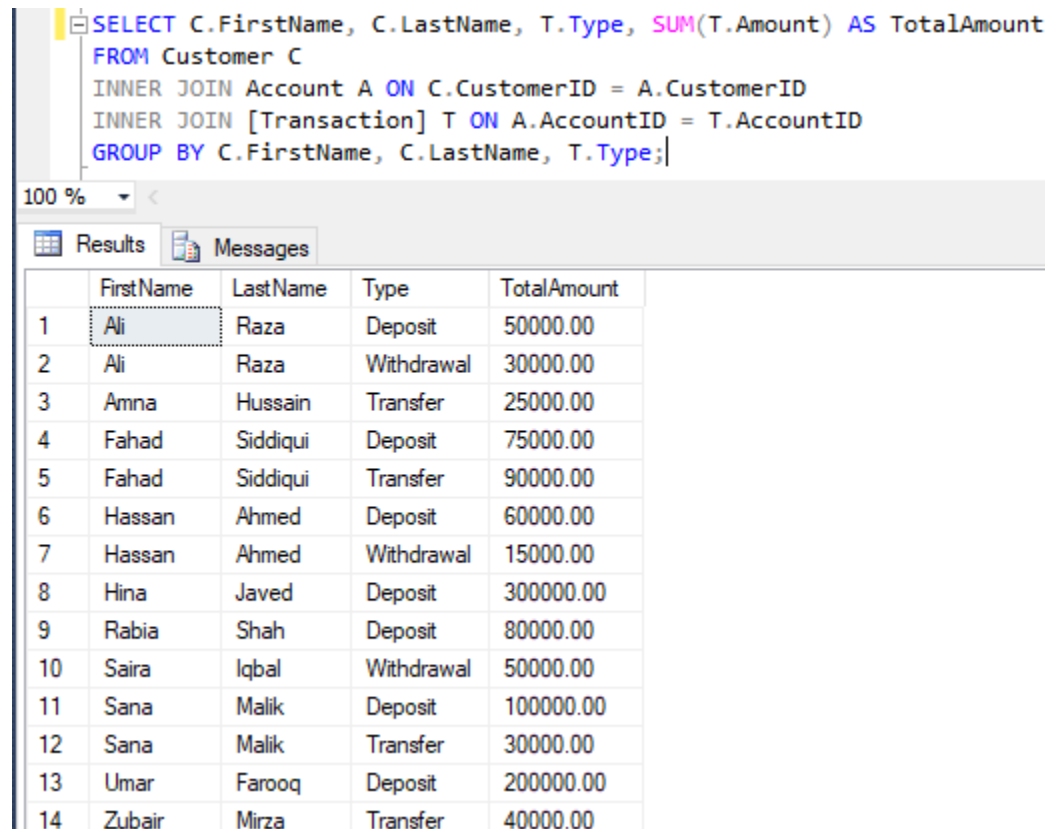
4. Information Retrieval & Relational Algebra

- Write 3 SQL queries using JOINS across multiple tables.
- At least 2 queries must include aggregate functions with GROUP BY.
- Provide corresponding Relational Algebra expressions for each query.

Query 1:

Total amount deposited/withdrawn per customer

- (JOIN + Aggregate + GROUP BY)



```
SELECT C.FirstName, C.LastName, T.Type, SUM(T.Amount) AS TotalAmount
FROM Customer C
INNER JOIN Account A ON C.CustomerID = A.CustomerID
INNER JOIN [Transaction] T ON A.AccountID = T.AccountID
GROUP BY C.FirstName, C.LastName, T.Type;
```

	FirstName	LastName	Type	TotalAmount
1	Ali	Raza	Deposit	50000.00
2	Ali	Raza	Withdrawal	30000.00
3	Amna	Hussain	Transfer	25000.00
4	Fahad	Siddiqui	Deposit	75000.00
5	Fahad	Siddiqui	Transfer	90000.00
6	Hassan	Ahmed	Deposit	60000.00
7	Hassan	Ahmed	Withdrawal	15000.00
8	Hina	Javed	Deposit	300000.00
9	Rabia	Shah	Deposit	80000.00
10	Saira	Iqbal	Withdrawal	50000.00
11	Sana	Malik	Deposit	100000.00
12	Sana	Malik	Transfer	30000.00
13	Umar	Farooq	Deposit	200000.00
14	Zubair	Mirza	Transfer	40000.00

Expression:

$\gamma_{\text{FirstName, LastName, Type, SUM(Amount)}} ($
 $(\text{Customer} \bowtie \text{Customer.CustomerID} = \text{Account.CustomerID} \text{ Account})$
 $\bowtie \text{Account.AccountID} = \text{Transaction.AccountID} \text{ Transaction}$
 $)$

Query 2:

Joins Customer and Account, then shows total account balance held by customers in each city.

- Total balance per city (JOIN + Aggregate + GROUP BY)

```
SELECT C.City, SUM(A.Balance) AS TotalBalance
FROM Customer C
INNER JOIN Account A ON C.CustomerID = A.CustomerID
GROUP BY C.City;
```

100 % <

Results Messages

	City	TotalBalance
1	Islamabad	860000.00
2	Karachi	415000.00
3	Lahore	1650000.00
4	Peshawar	455000.00
5	Rawalpindi	75000.00

Expression:

γ City, SUM(Balance) (
Customer $\bowtie$ Customer.CustomerID = Account.CustomerID Account
)

Query 3:

Branches where total approved loan amount exceeds 1,000,000

- JOIN across 3 tables + COUNT + SUM + GROUP BY + HAVING

```
SELECT B.BranchName,B.City,COUNT(L.LoanID)AS LoanCount, SUM(L.ApprovedAmount) AS TotalApproved
FROM Loan L
INNER JOIN Employee E ON L.ApprovedByEmpID = E.EmployeeID
INNER JOIN Branch B ON E.BranchID = B.BranchID
WHERE L.LoanStatus = 'Approved'
GROUP BY B.BranchName, B.City
HAVING SUM(L.ApprovedAmount) > 1000000;
```

100 %

Results Messages

	BranchName	City	LoanCount	TotalApproved
1	Blue Area Branch	Islamabad	3	2700000.00
2	Gulberg Branch	Lahore	3	9600000.00
3	Cannt Branch	Peshawar	2	2800000.00

Expression:

$\sigma_{\text{SUM}(\text{ApprovedAmount}) > 1000000} ($
 $\gamma_{\text{BranchName, City, COUNT(LoanID), SUM(ApprovedAmount)}} ($
 $\sigma_{\text{LoanStatus} = \text{'Approved'}} ($
 $(\text{Loan} \bowtie_{\text{Loan.ApprovedByEmpID} = \text{Employee.EmployeeID}} \text{Employee})$
 $\bowtie_{\text{Employee.BranchID} = \text{Branch.BranchID}} \text{Branch}$
)
)
)